

Neural Network and Fuzzy System

1. Difference between biological neuron and artificial neuron.
2. What is Artificial Neuron?
3. Explain the structure/Working of ANN.
4. Difference between Shallow Neural Network and Deep Neural Network.
5. Draw and explain the basic architecture of an Artificial Neural Network (ANN) showing the input, hidden, and output layers. How is the number of neurons in the input and output layers determined?
6. Effect of hidden layer/neuron numbers.
7. Design a simple Artificial Neural Network (ANN) that can recognize handwritten digits from 0 to 9.
8. Design a simple Artificial Neural Network (ANN) that can recognize handwritten digits from 0 to 9. Input Image (56x56)
9. What is overfitting ?
10. Why Generally nonlinear functions are used as activation functions.
11. Differentiate between regression and classification.
12. What is an activation function? Explain Sigmoid, ReLU, and Softmax activation functions and mention their uses.
13. What is a loss function? Name suitable loss functions for regression and classification problems
14. Explain The gradient descent algorithm
15. Overfitting Underfitting ,How to avoid overfitting
16. Forward propagation computation graph, backpropagation equation.
17. Backpropagation Algorithm math. (2 types).
18. Explain Nearest neighbour Classification Method.
19. Maths related to distance calculation and nearest neighbour classification.
20. pattern recognition and the two main stages of a pattern recognition system.